

DATABASE SYSTEMS LAB

Semester Project — Milestone 1

Student Fee Management System (SFMS)

ERD + Relational Database Schema

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Field	Details
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Program & Group	BS Computer Science — [A]
Project Title	Student Fee Management System (SFMS)

Milestones	Date	Version	Remarks
ERD diagram and schema	26-4-2026	1.0	ERD and relational schema prepared and submitted on 26-04-2026

1. Entity-Relationship Diagram (ERD)

The diagram below shows all five entities of the SFMS, their attributes, and the relationships between them using standard crow's-foot notation. Primary keys (PK) uniquely identify each record; Foreign keys (FK) link related tables.

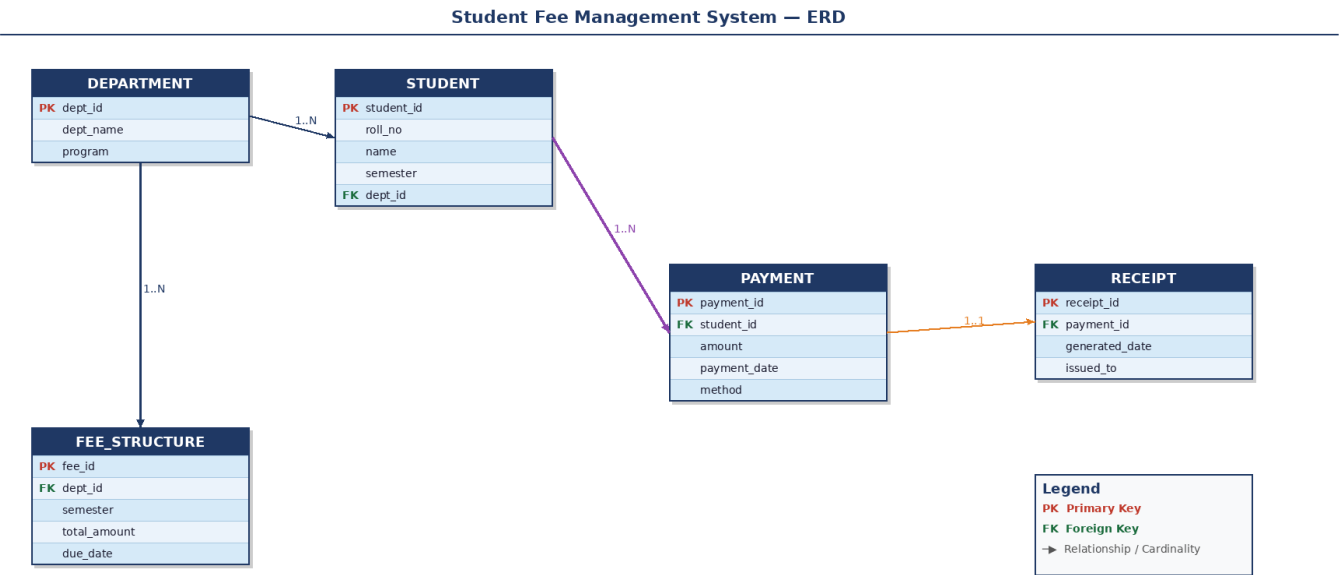


Figure 1 — ERD of the Student Fee Management System

1.1 Relationships

Parent Entity	Child Entity	Cardinality	Description
DEPARTMENT	STUDENT	One-to-Many (1..N)	A department has many students; each student belongs to one department
DEPARTMENT	FEE_STRUCTURE	One-to-Many (1..N)	A department defines fee structures for many semesters
STUDENT	PAYMENT	One-to-Many (1..N)	A student can make multiple payments; each payment belongs to one student

Parent Entity	Child Entity	Cardinality	Description
PAYMENT	RECEIPT	One-to-One (1..1)	Each payment generates exactly one receipt

2. Relational Database Schema

The schema consists of five tables in Third Normal Form (3NF). Each table is described below with its attributes, data types, and integrity constraints.

2.1 DEPARTMENT

Stores department and program information. Each department can have many students and multiple fee structures.

Attribute	Data Type	Key	Constraint / Description
dept_id	INT	PK	Auto-increment primary key
dept_name	VARCHAR(100)		Full name of the department (NOT NULL)
program	VARCHAR(100)		Degree program offered (e.g., BS CS)

2.2 STUDENT

Holds all registered student profiles. Each student belongs to one department.

Attribute	Data Type	Key	Constraint / Description
student_id	INT	PK	Auto-increment primary key
roll_no	VARCHAR(20)		Unique student roll number (NOT NULL, UNIQUE)
name	VARCHAR(100)		Full name of the student (NOT NULL)
semester	INT		Current semester (CHECK: 1–8)
dept_id	INT	FK	References DEPARTMENT(dept_id) — ON DELETE RESTRICT

2.3 FEE_STRUCTURE

Defines the total fee per department per semester, along with the payment due date.

Attribute	Data Type	Key	Constraint / Description
fee_id	INT	PK	Auto-increment primary key
dept_id	INT	FK	References DEPARTMENT(dept_id)

Attribute	Data Type	Key	Constraint / Description
semester	INT		Semester number (CHECK: 1–8, NOT NULL)
total_amount	DECIMAL(10,2)		Total fee for the semester (CHECK: > 0)
due_date	DATE		Last date for fee payment (NOT NULL)

2.4 PAYMENT

Records every fee payment transaction made by a student.

Attribute	Data Type	Key	Constraint / Description
payment_id	INT	PK	Auto-increment primary key
student_id	INT	FK	References STUDENT(student_id) — ON DELETE CASCADE
amount	DECIMAL(10,2)		Amount paid (CHECK: > 0, NOT NULL)
payment_date	DATE		Date payment was made (NOT NULL)
method	VARCHAR(20)		Payment mode: Cash or Bank (NOT NULL)

2.5 RECEIPT

Auto-generated receipt linked one-to-one with each confirmed payment.

Attribute	Data Type	Key	Constraint / Description
receipt_id	INT	PK	Auto-increment primary key
payment_id	INT	FK	References PAYMENT(payment_id) — ON DELETE CASCADE
generated_date	DATETIME		Timestamp of receipt generation (NOT NULL)
issued_to	VARCHAR(100)		Student name printed on receipt

3. Integrity Constraints Summary

The following constraints are enforced at the database level to guarantee data accuracy and consistency.

Constraint	Applied On	Purpose
PRIMARY KEY	All tables	Uniquely identifies every record; auto-incremented integer

Constraint	Applied On	Purpose
FOREIGN KEY	dept_id, student_id, payment_id	Enforces referential integrity between related tables
NOT NULL	Names, amounts, dates, methods	Prevents incomplete or missing data in critical fields
UNIQUE	roll_no (STUDENT)	Ensures no two students share the same roll number
CHECK	semester (1–8), amount (> 0)	Validates domain ranges at the database level